Oliver's Physics C E&M Cram Study Guide

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Contents

Chapter 1

Multivariable Calculus

In order to adequately grasp the concepts of electricity and magnetism, you should at least know the basics of multivariable calculus. Throughout the text, I assume knowledge of the rudiments of single-variable calculus. We begin with the **partial derivative**, which is the natural analog for the regular derivative for functions of multiple variables; given some function $z = f(x, y)$, we have that

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \tag{1.1}$$

$$\frac{\partial f}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y} \tag{1.2}$$

To take the partial derivative of a function, simply take the derivative with respect to the appropriate variable while "pretending" as if the other variables are constants. For instance, we have that

$$\frac{\partial}{\partial x}(3x^2 y) = 3y \frac{\partial}{\partial x}(x^2) = 6xy \tag{1.3}$$

The **dot product** of two vectors produces a scalar, like so:

$$\langle v_1, v_2, v_3\rangle \cdot \langle w_1, w_2, w_3\rangle = v_1w_1 + v_2w_2 + v_3w_3 \tag{1.4}$$

Note that the dot product of two parallel vectors (in the same direction) will be the lengths of both vectors multiplied by each other. The dot product of two perpendicular vectors will be zero. Meanwhile, the **cross product** of two vectors produces a vector perpendicular to both, like so:

$$\langle v_1, v_2, v_3\rangle \times \langle w_1, w_2, w_3\rangle = \langle v_2w_3 - v_3w_2, v_3w_1 - v_1w_3, v_1w_2 - v_2w_1\rangle \tag{1.5}$$

The cross product of two parallel vectors will be the zero vector.

The **gradient** of a (scalar) function $f(x, y, z)$ is equal to

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \tag{1.6}$$

The **divergence** of $\mathbf{F}(x, y, z) = \langle f_1(x, y, z), f_2(x, y, z), f_3(x, y, z)\rangle$ (a *vector field*) is

$$\nabla \cdot \mathbf{F}(x, y, z) = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \tag{1.7}$$

The **curl** of the same vector field can be calculated like so:

$$\nabla \times \mathbf{F}(x, y, z) = \left\langle \frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}, \frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}, \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right\rangle \tag{1.8}$$

You have likely integrated along an interval on the real line. Perhaps you have even calculated double integrals or triple integrals. But let's talk about **line integrals**. Line integrals integrate over a piecewise smooth curve, often denoted C. Suppose we have a curve C given by $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$, such that $a \leq t \leq b$. Then we can integrate a scalar function like so:

$$\int_C f(x,y,z)ds = \int_a^b f(x(t),y(t),z(t))\sqrt{[x'(t)]^2+[y'(t)]^2+[z'(t)]^2}dt \qquad (1.9)$$

Calculating the line integral of the function $f(x,y,z) = 1$ over a curve gives the arc length of the curve in question. Now, suppose that we have a vector field $\mathbf{F}(x,y,z)$. Then, we can integrate that over a curve as well:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(x(t),y(t),z(t)) \cdot \mathbf{r}'(t)dt \qquad (1.10)$$

If $\mathbf{F}(x,y,z) = \langle f_1(x,y,z), f_2(x,y,z), f_3(x,y,z)\rangle$, the line integral is sometimes written like

$$\int f_1dx + f_2dy + f_3dz \qquad (1.11)$$

It's the same thing.

Theorem 1 (Fundamental Theorem of Line Integrals)**.** *Suppose we have a curve* $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$ *such that* $a \leq t \leq b$*. If* $\mathbf{F}(x,y,z)$ *is* ***conservative*** *(that is,* $\mathbf{F} = \nabla f$ *for some function* f*) and its constituent functions are continuous, then*

$$\int_C \mathbf{F} \cdot d\boldsymbol{r} = \int_C \nabla f \cdot d\mathbf{r} = f(x(b),y(b),z(b)) - f(x(a),y(a),z(a)) \qquad (1.12)$$

Suppose we have a closed curve; that is, a curve that begins and ends at the same point. Then we write the line integral over such a curve as

$$\oint \mathbf{F} \cdot d\mathbf{r} \qquad (1.13)$$

As it turns out, this expression is always going to be 0 if and only if $\mathbf{F}$ is conservative.

Line integrals are not the only type of integral in vector calculus; indeed, you may encounter **surface integrals** as well. Suppose we have a surface S defined by $z = g(x, y)$ and R is its projection onto the xy-plane. The surface integral of a function f over S is

$$\iint_S f(x,y,z)dS = \iint_R f(x,y,g(x,y))\sqrt{1+\left(\frac{\partial g}{\partial x}\right)^2+\left(\frac{\partial g}{\partial y}\right)^2}\,dA \tag{1.14}$$

You will need to know how to evaluate double integrals to make good use of this.

What is dS? In general, dS is given by the following formula, where $\mathbf{r}(u, v)$ is the equation for a parametric surface:

$$dS = ||\mathbf{r}_u(u,v) \times \mathbf{r}_v(u,v)||\, dA \tag{1.15}$$

In this case, where $\mathbf{r}(u, v) = \langle r_1(u, v), r_2(u, v), r_3(u, v)\rangle$ $\mathbf{r}_u$ is $\left\langle \frac{\partial r_1}{\partial u}, \frac{\partial r_2}{\partial u}, \frac{\partial r_3}{\partial u} \right\rangle$. $\mathbf{r}_v$ is defined analogously.

More generally, the surface integral can be written as

$$\iint_S f(x,y,z)dS = \iint_D f(x(u,v),y(u,v),z(u,v))dS \tag{1.16}$$

We also have **flux integrals**, which integrate a vector field over a surface. These can be calculated like so:

$$\iint_S \mathbf{F}\cdot\mathbf{N}dS = \iint_D \mathbf{F}\cdot(\mathbf{r}_u \times \mathbf{r}_v)dA \tag{1.17}$$

$\mathbf{N}$ is considered to be a unit normal vector perpendicular to the surface. Note that there are often two choices of normal vectors; this phenomenon is referred to as **orientation** of a surface.

We shall now explore two important theorems in multivariable calculus: the

divergence theorem and **Stokes's theorem**.

Theorem 2 (Divergence Theorem)**.** *We have that*

$$\oiint_S \mathbf{F} \cdot \mathbf{N} dS = \iiint_Q \nabla \cdot \mathbf{F} dV \tag{1.18}$$

where Q is a solid region bounded by a closed surface S.

Theorem 3 (Stokes's Theorem)**.** *We have that*

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} dS \tag{1.19}$$

where S is an oriented surface bounded by a piecewise smooth simple closed curve C.

Bonus: Differential Forms

Note that these bonus sections will contain material significantly more difficult than that featured on the AP exam; do not be discouraged if you do not understand all of the content in this section.

To elegantly unify many of the theorems in multivariable calculus, we look toward the use of **differential forms**, which are multilinear and alternating functionals that take in vector values. That is to say, differential forms, which are written as ω often, take in some amount of vector values, like so: $\omega(X_1, \cdots X_k)$, and:

1. ω is **multilinear**:

$$\omega(X_1, \cdots, X_{i-1}, aX_i + bX_i', X_{i+1}, \cdots, X_k)$$

$$= a\omega(X_1, \cdots, X_i, \cdots X_k) + b\omega(X_1, \cdots, X_i', \cdots X_k) \tag{1.20}$$

2. ω is **alternating**: if we swap the values of X_i and X_j, the sign of ω changes.

We often write differential forms as a linear combination of 1-forms:

$$\omega = \sum_I f_I(x) dx_{i_1} \wedge \cdots \wedge dx_{i_k} \tag{1.21}$$

We also have that

$$dx_i \left(\frac{\partial}{\partial x_j} \right) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \tag{1.22}$$

Yes, $\frac{\partial}{\partial x_j}$ is a vector; it is called a **tangent vector**. And, the funky-looking δ_{ij} symbol is called a **Kronecker delta**.

There are certain operations on differential forms which may be performed. We can turn a k-form and an l-form into a $k + l$-form through the means of the **exterior product**, or **wedge product**; given $\omega = \sum_I f_I(x) dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ and $\eta = \sum_J g_J(x) dx_{j_1} \wedge \cdots \wedge dx_{j_l}$, we have

$$\omega \wedge \eta = \sum_{I,J} f_I g_J dx_{i_1} \wedge \cdots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \cdots \wedge dx_{j_l} \tag{1.23}$$

The **exterior derivative** d takes a k-form and turns it into a $k+1$-form; given ω as defined before,

$$d\omega = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(x) dx_j \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k} \tag{1.24}$$

Note that $dx_i \wedge dx_i = 0$.

Now we come to one of the crown jewels of calculus; **Stokes's theorem on manifolds**. This unifies the fundamental theorem of calculus, Stokes's theorem, and the divergence theorem. Note that a **manifold** is simply a generalization of a curve, a surface, a three-dimensional region (a more rigorous definition can be found in many differential geometry books). Observe:

Theorem 4 (Stokes's Theorem on Manifolds)**.** *Suppose M is an oriented n-dimensional manifold. Let ω be an $(n-1)$-form on M. Then*

$$\int_M d\omega = \int_{\partial M} \omega \tag{1.25}$$

*where ∂M is the **boundary** of M.*

One can see how this is very similar to that of the divergence and Stokes's theorems.

Finally, the **Hodge star operator** turns a k-form into an $n-k$-form, where n is the dimension of the manifold that the differential form pertains to. We set

$$*(dx_1 \wedge \cdots \wedge dx_k) = dx_{k+1} \wedge \cdots \wedge dx_n \tag{1.26}$$

Furthermore,

$$*1 = dx_1 \wedge \cdots \wedge dx_n, \quad *(dx_1 \wedge \cdots \wedge dx_n) = 1 \tag{1.27}$$

The Hodge star operator has a couple properties:

$$*(f\omega + g\eta) = f * \omega + g * \eta \tag{1.28}$$

$$* * \omega = (-1)^{k(n-k)}\omega \tag{1.29}$$

We can also define a generalized version of the **divergence** using the Hodge star operator and the exterior derivative:

$$\text{div } X = *d * \omega_X \tag{1.30}$$

where ω_X is the 1-form "corresponding" to X (to make this rigorous would require a degree of mathematical maturity beyond appropriate for that of a brief overview).

Chapter 2

Electric Fields

We are first going to study **electrostatics**, which concern the interactions between electric charges that are not moving. Two positive charges repel, and so do two negative charges, but a positive charge and a negative charge attract. Atoms have three particles: the **electron**, a negatively charged particle, the **proton**, which is positively charged, and the **neutron**, which has no charge. Protons and neutrons make up the **nucleus**, which is the core of the atom. The number of protons in an element is called the **atomic number** of the element. If electrons are removed from the atom, the atom becomes a **positive ion**; if electrons are added, the atom becomes a **negative ion**. The process of removing or adding electrons is called **ionization**. Note that *the algebraic sum of all the electric charges in any closed system must be constant*; that is, electric charge is conserved. Additionally, the magnitude of charge of the electron or proton is a natural unit of charge; every amount of electric charge that is observable must be a multiple of this unit.

A **conductor** allows for electric charge to move easily from one part of the material to another. An **insulator** is precisely the opposite. It is possible to charge by **induction**; that is, when you bring a charged object close to an uncharged conductor without touching it, the electrons in the conductor begin to move based on whether they are attracted or repelled by the charged object. The resulting charges that can be found in different parts of the conductor are called **induced charges**.

Let us discuss **Coulomb's law** for point charges. The magnitude of the electric force between two point charges is proportional to the product of the charges and inversely proportional to the square of the distance between them:

$$F = \frac{1}{4\pi\epsilon_0}\frac{|q_1 q_2|}{r^2} \tag{2.1}$$

The **coulomb** is the fundamental unit of charge, and the magnitude of the charge of the electron or proton is equal to $e \approx 1.60 \times 10^{-19}\mathrm{C}$. ϵ_0 is called the **electric constant**, and is approximately equal to $8.854 \times 10^{-12}\mathrm{C}^2/\mathrm{N}\cdot\mathrm{m}^2$. The **principle of superposition of forces** states that, if we have two charges acting on a third charge, the force acting on the charge is the vector sum of the forces of the two charges individually.

Let us now consider the electric field; charged objects create an **electric field**, which exerts a force on another charged object inside the field. To find out information about the electric field, we must employ a **test charge**. The electric field is the force on a test charge divided by the value of the test charge:

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}_0}{q_0} \tag{2.2}$$

Note that this formula is for point test charges only. The location of the charge producing the electric field is called the **source point**, and the point where we are examining the field is known as the **field point**. The electric field due to a point charge can be written like so:

$$\vec{\mathbf{E}} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}} \tag{2.3}$$

where q is the point charge, and $\hat{\mathbf{r}}$ is the unit vector from the point charge to the field point. $\vec{\mathbf{E}}$ is an example of a vector field.

Let's talk about how to calculate the electric field. These can be tricky, so it is important to think of a strategy while calculating.

1. Draw the situation, with charges and coordinate axes.

2. Make sure to note where the field point is as well.

3. Remember to use vector addition to apply the principle of superposition.

4. Be sure to use symmetry whenever possible.

5. You will most likely have to do an integral when calculating the electric field. To set up the integral, consider writing dE in terms of dQ, and then writing dQ in terms of dx, dy, or ds. Bear in mind that you may have to calculate the components of the electric field individually; that is, E_x, E_y, E_z.

We also have **electric field lines**, which are imaginary lines drawn such that they are tangent to the electric field vector at any given point. Be aware that electric field lines are not necessarily trajectories of a charged particle. Electric field lines emerge outward from positive charges and go to negative charges.

A pair of point charges with equal magnitude and opposite sign separated by some distance is called an **electric dipole**. The torque on an electric dipole is given by the formula

$$\tau = (qE)(d\sin\phi) \tag{2.4}$$

where d is the distance between the two charges and ϕ is the angle between the electric field and the dipole axis. The magnitude of the **electric dipole moment** is the product of the charge and the separation distance:

$$p = qd \tag{2.5}$$

whereas its direction is along the dipole axis from the negative to the positive charge. Therefore, we have that

$$\tau = pE \sin \phi \tag{2.6}$$

We can therefore write the torque in vector form as

$$\vec{\tau} = \vec{\mathbf{p}} \times \vec{\mathbf{E}} \tag{2.7}$$

Furthermore, to find the potential energy of an electric dipole, we integrate the negative of the torque (as the torque is in the direction of decreasing ϕ):

$$dW = \tau d\phi = -pE \sin \phi d\phi \implies W = \int_{\phi_1}^{\phi_2} (-pE \sin \phi) d\phi \tag{2.8}$$

Since $W = U_1 - U_2$, we naturally obtain the definition of potential energy in this system:

$$U(\phi) = -pE \cos \phi \implies U = -\vec{\mathbf{p}} \cdot \vec{\mathbf{E}} \tag{2.9}$$

Let's talk about **electric flux**. Electric flux is defined as

$$\Phi_E = \iint E \cos \phi dA = \iint E_\perp dA = \iint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} \tag{2.10}$$

where ϕ is the angle between the electric field and a normal vector (perpendicular to the surface), and $E_\perp$ is the component of the electric field perpendicular to the surface. Note that for a uniform electric field and a flat surface, we have

$$\Phi_E = EA\cos\phi = E_{\perp}A = \vec{\mathbf{E}} \cdot \vec{\mathbf{A}} \tag{2.11}$$

where $\vec{\mathbf{A}}$ is the area times the unit normal vector perpendicular to the surface. Note that, when it comes to calculating electric flux, if the magnitude of the electric field is the same throughout the surface and the electric field is perpendicular to the surface, it does not depend on where on the surface you are, so you can take the electric field out of the integral $\Phi_E = E \iint dA$. This is not necessarily the case for every electric flux calculation, though. Also, $\iint dA$ is just the surface area of the surface.

Let's talk about Gauss's law:

$$\Phi_E = \oiint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{Q_{\text{encl}}}{\epsilon_0} \tag{2.12}$$

In other words, the total electric flux through a *closed* surface is equal to the total electric charge inside the surface divided by ϵ_0. One important thing to note when performing these calculations is that if you come across a conductor, under electrostatic conditions, the charge on a conductor must reside on the conductor's surface. For example, if we have a sphere with charge +3C, then all that positive charge is on the surface of the sphere. In this exam, you will need to use Gauss's law often. Gauss's law is effective if the surface has spherical, cylindrical, or planar symmetry; I doubt that the exam will require surfaces with no such symmetry.

1. List known and unknown quantities and identify which variable you are trying to figure out. In many cases, this is a great first step.

2. Find a good Gaussian surface to integrate over. This could be a sphere or a cylinder, usually.

3. Evaluate the integral $\oiint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}}$. Recall that if the electric field is parallel to the surface at a point, there is no contribution to the integral there. If the electric field is uniform and perpendicular to a part of the surface, you can take the magnitude of the electric field there and multiply it by the appropriate area.

4. If there is no charge within a Gaussian surface, the electric field may not always be zero, but the total electric flux through the surface will be zero.

5. The flux integral can be thought of loosely as the difference between the numbers of electric lines of force leaving versus entering the Gaussian surface; this should give the sign of the charge, but not the exact charge itself. You will only be able to determine if there is a negative enclosed charge, positive enclosed charge, or no enclosed charge.

Gauss's law implies that we are able to find a formula for the electric field at the surface of a conductor:

$$E_\perp = \frac{\sigma}{\epsilon_0} \tag{2.13}$$

where σ is the surface charge density.

Bonus: The Divergence and Curl of $\vec{\mathbf{E}}$

Gauss's law in differential form is written as follows:

$$\nabla \cdot \vec{\mathbf{E}} = \frac{\rho}{\epsilon_0} \tag{2.14}$$

where ρ is the charge density. This is the divergence of $\vec{\mathbf{E}}$. To calculate the curl of $\vec{\mathbf{E}}$, we consider that the integral around a closed path of the electric field is zero:

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = 0 \tag{2.15}$$

so, applying Stokes's theorem, we get

$$\nabla \times \vec{\mathbf{E}} = \mathbf{0} \tag{2.16}$$

Chapter 3

Electric Potential

We begin by calculating the **electric potential energy** of two point charges. This can be done by recognizing that

$$W_{a\to b} = \int_{r_a}^{r_b} F\cos\phi dl = U_a - U_b \tag{3.1}$$

and

$$F_r = \frac{1}{4\pi\epsilon_0}\frac{qq_0}{r^2} \tag{3.2}$$

We have

$$U = \frac{1}{4\pi\epsilon_0}\frac{qq_0}{r} \tag{3.3}$$

When it comes to calculating the electric potential energy with several point charges, we have

$$U = \frac{q_0}{4\pi\epsilon_0}\sum_i \frac{q_i}{r_i} \tag{3.4}$$

The potential energy required to assemble the point charges in that way can also be calculated; namely, it is

$$U = \frac{1}{4\pi\epsilon_0} \sum_{i<j} \frac{q_i q_j}{r_{ij}} \tag{3.5}$$

The potential energy difference $U_a - U_b$ can be thought of as the work that is done by the electric force when the particle moves from a to b, and can also be thought of as the work that must be done by an external force to move the particle from b to a against the electric force.

Potential is potential energy per unit charge, and the SI unit of potential is the **volt**. V_{ab} is the work done by the electric force when a unit charge moves from a to b, and is also the work that must be done to move a unit charge slowly from b to a against the electric force. The voltmeter measures the difference of potential between two points. We can calculate the electric potential due to a point charge:

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \tag{3.6}$$

Similarly, we calculate the electric potential due to multiple point charges:

$$V = \frac{1}{4\pi\epsilon_0} \sum_{i} \frac{q_i}{r_i} \tag{3.7}$$

Finally, to calculate the electric potential due to a continuous distribution of charge, we have

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r} \tag{3.8}$$

To calculate $V_a - V_b$, we have the line integral

$$V_a - V_b = \int_a^b \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} \tag{3.9}$$

The **electron volt** eV is a unit of energy, specifically the energy required for a particle with charge e to move through a potential difference of 1 volt. It is 1.602×10^{-19}J.

To calculate electric potential, we employ the following strategy:

1. Draw the charges and coordinate axes.
2. Label the point where you want to calculate the electric potential V.
3. To calculate the electric potential from several point charges, use the relevant equation. Calculating the electric potential from a continuous charge distribution is harder.

If you can calculate the potential from a very long (uniformly distributed) charged line from some distance, the potential from a ring with uniformly distributed charge, and the potential from a (uniformly distributed) charged line of finite length, then you should be good to go.

An **equipotential surface** is a surface on which the electric potential V is equal at every point. Note that field lines and equipotential surfaces are always mutually perpendicular. When all charges are at rest, the surface of a conductor must always be an equipotential surface. Furthermore, the electric field just outside a conductor must be perpendicular to the surface at every point. Also, the entire solid volume of a conductor is at the same potential. This is all under electrostatic conditions.

Furthermore, we have that

$$\vec{\mathbf{E}} = -\nabla V \tag{3.10}$$

Additionally, if you are instead considering the radial component of the electric field, we have

$$E_r = -\frac{\partial V}{\partial r} \tag{3.11}$$

Bonus: Laplace's Equation

Poisson's equation is as follows:

$$\nabla^2 V = -\frac{1}{\epsilon_0}\rho \tag{3.12}$$

Often we are interested in regions where $\rho = 0$. Then, Poisson's equation becomes **Laplace's equation**:

$$\nabla^2 V = 0 \tag{3.13}$$

It may be important to note that ∇^2, sometimes written as Δ, is known as the Laplacian, and is the divergence of the gradient of a function: $\nabla^2 f = \nabla \cdot (\nabla f)$. Solutions of Laplace's equation are called **harmonic functions**. In one dimension, Laplace's equation becomes

$$\frac{d^2V}{dx^2} = 0 \tag{3.14}$$

Of course, the solutions to this equation are in the form of $V(x) = mx + b$. This gives rise to some interesting properties. First, $V(x)$ is the average of $V(x+a)$ and $V(x-a)$; you can verify this yourself. Additionally, maxima and minima of V must occur at the endpoints, not anywhere in the middle.

In two dimensions, Laplace's equation becomes

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0 \tag{3.15}$$

Once again, the value of V at a point (x, y) is, interestingly enough, the average of those around the point. That is to say,

$$V(x,y) = \frac{1}{2\pi R} \oint_{\text{circle}} V dl \tag{3.16}$$

where R is the radius of the circle, which is centered at (x, y). Additionally, all extrema still occur at the boundaries, just like in one dimension.

In three dimensions, we again have that the value of V at a point $\mathbf{r}$ is the average value of V over a spherical surface with radius R centered at that point:

$$V(\mathbf{r}) = \frac{1}{4\pi R^2} \oiint_{\text{sphere}} V dA \tag{3.17}$$

And once again, extrema of V must occur at boundaries.

Earnshaw's theorem states that a charged particle cannot be held in a stable equilibrium by electrostatic forces alone.

Laplace's equation is very useful, but it does not alone determine the potential; we need things called **boundary conditions** to help us identify a unique V. As it turns out, there exist **uniqueness theorems** that will aid us in our efforts; here is the first uniqueness theorem:

Theorem 5 (First Uniqueness Theorem)**.** *The solution to Laplace's equation in some volume $\mathcal{V}$ is uniquely determined if V is specified on the boundary surface $\mathcal{S}$.*

Proof. Suppose for the sake of contradiction that there were two different solutions to Laplace's equation that are the same everywhere on the boundary surface. That is,

$$\nabla^2 V_1 = 0 \quad \nabla^2 V_2 = 0 \tag{3.18}$$

Consider the difference between the two: $V_3 \equiv V_1 - V_2$. This too obeys Laplace's equation, as

$$\nabla^2 V_3 = \nabla^2(V_1 - V_2) = \nabla^2 V_1 - \nabla^2 V_2 = 0 \tag{3.19}$$

and it takes the value zero on all boundaries, since V_1 and V_2 are equal there. Since it obeys Laplace's equation, and since it takes the value zero on all boundaries, recall that the maxima and minima are found on the boundaries. That means that the maximum and minimum of V_3 are both zero. So, V_3 must be zero everywhere, even inside the boundary surface. That implies that $V_1 = V_2$, and therefore there must be *one* unique solution to Laplace's equation based on the appropriate boundary conditions. $\square$

There also exists a second uniqueness theorem, stated as follows:

Theorem 6 (Second Uniqueness Theorem)**.** *In a volume $\mathcal{V}$ surrounded by conductors and containing a specified charge density ρ, the electric field is uniquely determined if the total charge on each conductor is given.*

Proof. Let us suppose for the sake of contradiction that there do exist two different electric fields that are determined based on these conditions. We must have that

$$\nabla \cdot \vec{\mathbf{E}}_1 = \frac{\rho}{\epsilon_0} \quad \text{and} \quad \nabla \cdot \vec{\mathbf{E}}_2 = \frac{\rho}{\epsilon_0} \tag{3.20}$$

and, for each conductor,

$$\oiint_{i\text{th conducting surface}} \vec{\mathbf{E}}_1 \cdot d\vec{\mathbf{A}} = \frac{Q_i}{\epsilon_0} \tag{3.21}$$

and

$$\oiint_{i\text{th conducting surface}} \vec{\mathbf{E}}_2 \cdot d\vec{\mathbf{A}} = \frac{Q_i}{\epsilon_0} \tag{3.22}$$

Meanwhile, on the outer boundary of the volume, we have

$$\oiint \vec{\mathbf{E}}_1 \cdot d\vec{\mathbf{A}} = \frac{Q_{\text{tot}}}{\epsilon_0} \quad \text{and} \quad \oiint \vec{\mathbf{E}}_2 \cdot d\vec{\mathbf{A}} = \frac{Q_{\text{tot}}}{\epsilon_0} \tag{3.23}$$

Consider the difference $\vec{\mathbf{E}}_3 = \vec{\mathbf{E}}_1 - \vec{\mathbf{E}}_2$. Of course,

$$\nabla \cdot \vec{\mathbf{E}}_3 = 0 \quad \text{and} \quad \oiint \vec{\mathbf{E}}_3 \cdot d\vec{\mathbf{A}} = 0 \tag{3.24}$$

Since each conductor is an equipotential, it follows that V_3 is a constant over each conducting surface. Consider the following identity:

$$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f) \tag{3.25}$$

We can use this rule to show that

$$\nabla \cdot (V_3\vec{\mathbf{E}}_3) = V_3(\nabla \cdot \vec{\mathbf{E}}_3) + \vec{\mathbf{E}}_3 \cdot (\nabla V_3) = V_3(0) + \vec{\mathbf{E}}_3 \cdot (-\vec{\mathbf{E}}_3) = -(E_3)^2 \tag{3.26}$$

Then, we use the divergence theorem to show that

$$-\iiint_{\mathcal{V}} (E_3)^2 dV = \iiint_{\mathcal{V}} \nabla \cdot (V_3\vec{\mathbf{E}}_3) dV = \oiint_{\mathcal{S}} V_3\vec{\mathbf{E}}_3 \cdot d\vec{\mathbf{A}} \tag{3.27}$$

But V_3 is a constant over each surface, and don't forget that

$$\oiint \vec{\mathbf{E}}_3 \cdot d\vec{\mathbf{A}} = 0 \tag{3.28}$$

so it turns out that

$$\oiint_{\mathcal{S}} V_3\vec{\mathbf{E}}_3 \cdot d\vec{\mathbf{A}} = 0 \implies -\iiint_{\mathcal{V}} (E_3)^2 dV = 0 \tag{3.29}$$

Bear in mind that the second integral is never negative (because of the squared term), so the only way that it can be zero is if E_3 is zero everywhere. Therefore, $\vec{\mathbf{E}}_1 = \vec{\mathbf{E}}_2$, and thus uniqueness is shown. $\square$

Chapter 4

Capacitors

Two conductors separated by an insulator are considered to be a **capacitor**. The **capacitance** of a capacitor is defined as

$$C = \frac{Q}{V_{ab}} \tag{4.1}$$

where Q is the magnitude of charge on each conductor, and V_{ab} is the potential difference between conductors, where a has charge $+Q$, and b has charge $-Q$. The SI unit of capacitance is the **farad**, where $1\text{F} = 1\text{C}/\text{V}$. The capacitance of a parallel-plate capacitor in a vacuum is

$$C = \epsilon_0 \frac{A}{d} \tag{4.2}$$

where A is the area of one plate, and d is the distance between plates.

If we have capacitors in series, the equivalent capacitance of all of them combined C_{eq} is equal to

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \tag{4.3}$$

In parallel, the equivalent capacitance is equal to

$$C_{\mathrm{eq}} = C_1 + C_2 + C_3 + \cdots \tag{4.4}$$

The potential energy stored in a capacitor can be calculated by the following:

$$dW = v dq = \frac{q dq}{C} \implies W = \int_0^W dW = \frac{1}{C}\int_0^Q q dq = \frac{Q^2}{2C} \tag{4.5}$$

$$\implies U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV \tag{4.6}$$

This works because we define the potential energy of an uncharged capacitor to be zero, so W is equal to U. The energy per unit volume in the space between the plates of a parallel-plate capacitor is called the **energy density** u, and it is

$$u = \frac{1}{2}\epsilon_0 E^2 \tag{4.7}$$

A nonconducting material is known as a **dielectric**. The original capacitance $C_0 = Q/V_0$, where V_0 is the potential difference in the capacitor with a vacuum between the conductors. The **dielectric constant** K of the dielectric material can be found by the following formula:

$$K = \frac{C}{C_0} \tag{4.8}$$

We also have, as a result,

$$V = \frac{V_0}{K} \quad \text{and} \quad E = \frac{E_0}{K} \tag{4.9}$$

The **permittivity** of the dielectric ϵ is known as

$$\epsilon = K\epsilon_0 \tag{4.10}$$

So, we have that the capacitance of a parallel-plate capacitor, with a dielectric between plates, is

$$C = KC_0 = K\epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d} \tag{4.11}$$

And the electric energy density in a dielectric is

$$u = \frac{1}{2} K\epsilon_0 E^2 = \frac{1}{2}\epsilon E^2 \tag{4.12}$$

We have that the electric constant ϵ_0 is also known as the **vacuum permittivity**.

Free charges are charges that are added to and removed from the conducting capacitor plates. We have that in a dielectric, Gauss's law becomes

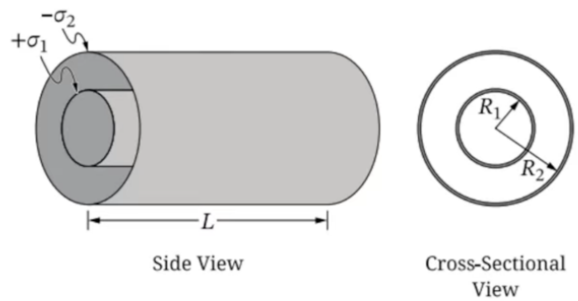
$$\oiint K\vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{Q_{\text{encl-free}}}{\epsilon_0} \tag{4.13}$$

where $Q_{\text{encl-free}}$ is the total free charge enclosed by the Gaussian surface.

When dielectric materials are exposed to particularly strong electric fields, dielectrics become conductors, in a process known as dielectric breakdown. The maximum field that a dielectric can withstand without undergoing dielectric breakdown is known as the material's dielectric strength.

Problem

An isolated, air-filled, charged capacitor consists of two conducting, coaxial, cylindrical shells that each have length L. The inner shell has radius R_1 and the outer shell has radius R_2, where $R_1 < R_2 \ll L$. The surface charge densities (amounts of charge per unit area) of the inner and outer shells are $+\sigma_1$ and $-\sigma_2$, respectively. The absolute values of the total charges on the shells are equal.



1. (a) Using Gauss's law, **derive** an expression for the magnitude E of the electric field as a function of the radial distance r from the center of the capacitor for the region $R_1 < r < R_2$. Express your answer in terms of R_1, σ_1, r, and physical constants, as appropriate.

   (b) **Derive** an expression for the absolute value $|\Delta V|$ of the potential difference between the outer and inner shells in terms of R_1, R_2, σ_1, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

   (c) **Sketch** a graph of E as a function of r from $r = 0$ to a position that is outside the outer shell.

2. A material of dielectric constant κ is inserted into the isolated, charged capacitor such that the material fills the region $R_1 < r < R_2$. **Derive** an expression for the capacitance C of the capacitor with the material inserted in terms of L, R_1, R_2, κ, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

Bonus: Polarization

When a neutral atom is placed in an electric field, the atom is **polarized**. The positive nucleus may tend slightly toward the direction of the field, and the negative electron cloud may tend slightly against it. In extreme cases, this results in ionization, but in less extreme cases, the atom reaches a sort of equilibrium state where it is polarized. There exists a dipole moment where

$$\vec{\mathbf{p}} = \alpha \vec{\mathbf{E}} \tag{4.14}$$

α is called **atomic polarizability**. Furthermore, for some molecules, the field must be split up into parallel and perpendicular components, and therefore the dipole moment must be calculated like so:

$$\vec{\mathbf{p}} = \alpha_{\perp} \vec{\mathbf{E}}_{\perp} + \alpha_{\parallel} \vec{\mathbf{E}}_{\parallel} \tag{4.15}$$

For other molecules, you must account for all possible directions:

$$\begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = \begin{pmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} \tag{4.16}$$

The matrix α_{ij} is known as the **polarizability tensor**. What is a tensor? It is something that transforms like a tensor.

Let us examine the formula for the force on a dipole in a nonuniform field:

$$\vec{\mathbf{F}} = \vec{\mathbf{F}}_{+} + \vec{\mathbf{F}}_{-} = q(\vec{\mathbf{E}}_{+} - \vec{\mathbf{E}}_{-}) = q(\Delta \vec{\mathbf{E}}) \tag{4.17}$$

We have that

$$\Delta E_x \approx (\nabla E_x) \cdot \vec{\mathbf{d}} \tag{4.18}$$

and there exist analogous equations for E_y and E_z. We can write this as

$$\Delta \vec{\mathbf{E}} = (\vec{\mathbf{d}} \cdot \nabla)\vec{\mathbf{E}} \tag{4.19}$$

so

$$\vec{\mathbf{F}} = (\vec{\mathbf{p}} \cdot \nabla)\vec{\mathbf{E}} \tag{4.20}$$

Note that when there exist a lot of little dipoles pointing along the direction of the field, then the material becomes **polarized**. The dipole moment per unit volume is known as the **polarization**.

Chapter 5

Circuits

We begin with a treatment of current. **Current** is a motion of charge from one region to another; by convention, we have that current I is in the direction where there is a flow of *positive* charge. Furthermore, the *drift velocity* is the motion of the moving charged particles in the direction of the electric force $\hat{\mathbf{F}}$. The SI unit of current is the **ampere**, which is 1 coulomb per second. Let n be the **concentration** of particles. We have that

$$I = \frac{dQ}{dt} = n|q|v_d A \tag{5.1}$$

where v_d is the drift speed, A is the cross-sectional area, and q is the charge per particle. The vector current density is defined as

$$\hat{\mathbf{J}} = nq\vec{\mathbf{v}}_d \tag{5.2}$$

where $\vec{\mathbf{v}}_d$ is the drift velocity.

The **resistivity** of a material is defined as

$$\rho = \frac{E}{J} \tag{5.3}$$

where E is the magnitude of the electric field in the material, and J is the magnitude of the current density caused by the electric field. The reciproca of resistivity is **conductivity**. The resistivity of temperature T is defined as

$$\rho(T) = \rho_0[1 + \alpha(T - T_0)] \tag{5.4}$$

where ρ_0 is the resistivity at temperature T_0, and α is the **temperature coefficient of resistivity**.

The **resistance** of a conductor is defined as

$$R = \frac{\rho L}{A} \tag{5.5}$$

where L is length and A is cross-sectional area. Now we come to **Ohm's law**:

$$V = IR \tag{5.6}$$

The SI unit of resistance is the **ohm**, or one volt per ampere. A device made to have a specific value of resistance is called a **resistor**.

A conductor must be part of a **complete circuit** in order to have a steady current. The force that makes current flow from a lower to higher potential is called **electromotive force**, or emf. The **internal resistance** of the source r is the resistance from a source of emf. We have that

$$V_{ab} = \mathcal{E} - Ir \tag{5.7}$$

where $\mathcal{E}$ is the emf of the source. We also have

$$I = \frac{\mathcal{E}}{R + r} \tag{5.8}$$

from combining the previous equation with Ohm's law. Note that a battery is not a "current source." That is to say, it does not always produce the same current no matter what circuit. Depending on the resistance of the external circuit, the battery may produce more or less current. A **voltmeter** measures the potential difference between the terminals, and the **ammeter** measures the current passing through it.

Let us discuss **power**. Its formula is

$$P = V_{ab}I = I^2R = \frac{V_{ab}^2}{R} \tag{5.9}$$

We have that

$$V_{ab} = IR_{\text{eq}} \tag{5.10}$$

where V_{ab} is the potential difference between terminals a and b of the network, and I is the current at a or b. To find the equivalent resistance of resistors in series, we have

$$R_{\text{eq}} = R_1 + R_2 + R_3 + \cdots \tag{5.11}$$

In parallel, we have

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots \tag{5.12}$$

We will now consider the topic of **Kirchhoff's rules**. A **junction** is a point where three or more conductors meet. A **loop** is a closed conducting path. Kirchhoff's junction rule is that the sum of the currents into any junction equals zero; that is, an equal amount of current goes into a junction as it goes out.

$$\sum I = 0 \tag{5.13}$$

Kirchhoff's loop rule states that the sum of the potential differences around any loop equals zero:

$$\sum V = 0 \tag{5.14}$$

These will help immensely when it comes to finding the voltage and resistance of the resistors.

A circuit that has a resistor and capacitor in series is called an **R-C circuit**. We have that the capacitor charge of a charging R-C circuit is

$$q = C\mathcal{E}(1 - e^{-t/RC}) = Q_f(1 - e^{-t/RC}) \tag{5.15}$$

where Q_f is the final capacitor charge. The current can be found by

$$i = \frac{dq}{dt} = \frac{\mathcal{E}}{R}e^{-t/RC} = I_0 e^{-t/RC} \tag{5.16}$$

where I is the initial current. RC is the **time constant**, or the **relaxation time**, of the circuit, which measures how quickly the capacitor charges:

$$\tau = RC \tag{5.17}$$

Now let's discuss a discharging capacitor. The capacitor charge is

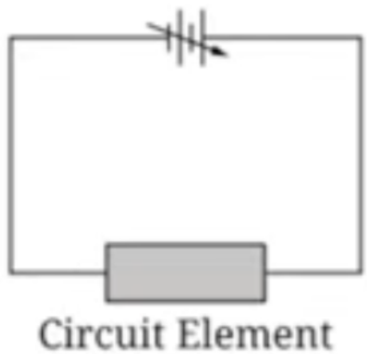
$$q = Q_0 e^{-t/RC} \tag{5.18}$$

and the current is

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC}e^{-t/RC} = I_0 e^{-t/RC} \tag{5.19}$$

Problem

In Experiment 1, students are asked to use a graph to determine the resistivity ρ_1 of a circuit element that is connected to a variable power supply. The circuit element is cylindrical and has uniform resistivity. The students have access to a voltmeter, an ammeter, and a ruler.



1. **Describe** a procedure for collecting data that would allow the students to use a graph to determine ρ_1, including any steps necessary to reduce experimental uncertainty.

2. **Describe** how the collected data could be graphed and how that graph would be analyzed to determine ρ_1.

In Experiment 2, the students are asked to use a graph to determine the resistivity ρ_2 of solid, cylindrical resistors made of the same material but of different lengths L. The cross-sectional area of each resistor is $5.0 \times 10^{-6}\text{m}^2$. The students directly measure the resistance R between the ends of each resistor.

L (m)	R (Ω)
0.010	0.90
0.020	1.6
0.030	2.5
0.040	3.2
0.050	4.0

1. **Indicate** two quantities, either measured quantities or additional calculated quantities, that could be graphed to produce a straight line that could be used to determine ρ_2.

2. Create a graph of the quantities indicated in part 1.

3. Using the best-fit line that you drew in part 2, **calculate** an experimental value for ρ_2.

Chapter 6

Magnetic Fields

You should know what a **magnet** is. As it turns out, electricity is very closely related to magnetism. A current creates a **magnetic field** in the surrounding space, and the magnetic field exerts a force on any other current present in the field. We have that the magnetic force on a moving charged particle can be described as

$$\vec{\mathbf{F}} = q\vec{\mathbf{v}} \times \vec{\mathbf{B}} \tag{6.1}$$

where $\vec{\mathbf{v}}$ is the velocity of the particle and $\vec{\mathbf{B}}$ is the magnetic field. Be sure to remember the **right-hand rule**. Index finger is in the direction of $\vec{\mathbf{v}}$, middle finger is in the direction of $\vec{\mathbf{B}}$, and the thumb will be in the direction of $\vec{\mathbf{F}}$. Some poor soul on the exam will use their left hand instead of their right hand (I am left handed, so I am particularly susceptible to this). **Do not be that guy!** The unit of B is the **tesla**, which is 1 N/A $\cdot$ m. The total force on a charged moving particle is the sum of the electric and magnetic forces:

$$\hat{\mathbf{F}} = q(\hat{\mathbf{E}} + \vec{\mathbf{v}} \times \vec{\mathbf{B}}) \tag{6.2}$$

We can represent magnetic fields with **magnetic field lines**, which emerge from the north pole and tend toward the south pole. The more densely the field lines are, the stronger the field is there. Once again, magnetic field lines are not necessarily force lines. In fact, they do not even point in the direction of the force on a charge. Let us discuss the **magnetic flux** through a surface, which is very similar to the electric flux:

$$\Phi_B = \iint B \cos\phi dA = \iint B_\perp dA = \iint \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} \tag{6.3}$$

There also exists Gauss's law for magnetism, which is very simple:

$$\oiint \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = 0 \tag{6.4}$$

The magnetic field $\vec{\mathbf{B}}$ is sometimes known as the **magnetic flux density**.

We have that motion of a charged particle under the action of a magnetic field *alone* is always motion with constant speed. Furthermore, we have

$$F = |q|vB = m\frac{v^2}{R} \tag{6.5}$$

so the radius of a circular orbit in a magnetic field is

$$R = \frac{mv}{|q|B} \tag{6.6}$$

where m is the mass of the particle, v is the speed of the particle, q is the charge of the particle, and B is the magnitude of the magnetic field. Furthermore, the angular speed ω is

$$\omega = \frac{v}{R} = v\frac{|q|B}{mv} = \frac{|q|B}{m} \tag{6.7}$$

The **cyclotron frequency** is $f = \omega/2\pi$. It is independent of the radius of the path.

Let us discuss the magnetic force on a straight wire segment:

$$\vec{\mathbf{F}} = I\vec{\mathbf{l}} \times \vec{\mathbf{B}} \tag{6.8}$$

where I is current, and $\vec{\mathbf{l}}$ is the vector length of the segment, pointing in the direction of the current. It is sometimes useful to use the following formula instead:

$$d\vec{\mathbf{F}} = Id\vec{\mathbf{l}} \times \vec{\mathbf{B}} \tag{6.9}$$

You may have to construct a line integral to properly use this equation. Be sure to make use of symmetries whenever possible; the integrals that the exam will give should usually turn out to be very simple.

Note that the net force on a current loop in a uniform magnetic field must be equal to zero, but the net torque may not always be. In fact, we have that

$$\tau = IBA\sin\phi \tag{6.10}$$

where ϕ is the angle between the normal vector to the loop plane and the field direction. We define the **magnetic dipole moment** or magnetic moment as

$$\mu = IA \tag{6.11}$$

An object that experiences a magnetic torque is called a **magnetic dipole**. Therefore, we have

$$\vec{\tau} = \vec{\mu} \times \vec{\mathbf{B}} \tag{6.12}$$

The potential energy for a magnetic dipole in a magnetic field is analogous to its electric version:

$$U = -\vec{\mu} \cdot \vec{\mathbf{B}} = -\mu B \cos\phi \tag{6.13}$$

The **solenoid** is a helical winding of wire. The magnetic moment of a solenoid with N turns is $\mu = NIA$, and therefore,

$$\tau = NIAB \sin\phi \tag{6.14}$$

The **Hall effect** is a potential difference perpendicular to the direction of current in a conductor, when the conductor is placed in a magnetic field. The Hall potential has the requirement that the electric field must just balance the magnetic field on a moving charge. We have

$$nq = \frac{-J_x B_y}{E_z} \tag{6.15}$$

where n is the concentration of mobile charge carriers, q is the charge per carrier, J_x is the current density in the x direction, B_y is the magnetic field, and E_z is the electrostatic field in the conductor.

We have that the magnetic field magnitude at a given point due to a moving charge is

$$B = \frac{\mu_0}{4\pi} \frac{|q| v \sin\phi}{r^2} \tag{6.16}$$

where ϕ is the angle between the velocity vector and the distance vector. μ_0 is known as the **magnetic constant**. The magnetic field due to a point charge with constant velocity is hence

$$\vec{\mathbf{B}} = \frac{\mu_0}{4\pi} \frac{q \vec{\mathbf{v}} \times \hat{\mathbf{r}}}{r^2} \tag{6.17}$$

Remember that $\hat{\mathbf{r}}$ is the unit vector pointing in the direction of the distance vector. We have that

$$\mu_0 = 1.26 \times 10^{-6} \mathrm{N} \cdot \mathrm{s}^2/\mathrm{C}^2 \tag{6.18}$$

Interestingly enough, the speed of light is

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \tag{6.19}$$

The total magnetic field caused by several moving charges is the vector sum of the fields caused by the individual charges. Note that we have

$$d\vec{\mathbf{B}} = \frac{\mu_0}{4\pi} \frac{I d\vec{\mathbf{l}} \times \hat{\mathbf{r}}}{r^2} \tag{6.20}$$

Now we approach the famous **Biot-Savart law**, which states

$$\vec{\mathbf{B}} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{\mathbf{l}} \times \hat{\mathbf{r}}}{r^2} \tag{6.21}$$

The magnetic field near a long, straight, current-carrying conductor is

$$B = \frac{\mu_0 I}{2\pi r} \tag{6.22}$$

where r is the distance from the conductor.

The magnetic force per unit length between two long, straight, parallel, current-carrying conductors is

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi r} \tag{6.23}$$

where r is the distance between conductors.

The magnetic field on the axis of a circular, current-carrying loop is

$$B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}} \tag{6.24}$$

where a is the radius of the loop. Meanwhile, the magnetic field at the center of N circular current-carrying loops is

$$B_x = \frac{\mu_0 N I}{2a} \tag{6.25}$$

where N is the number of loops.

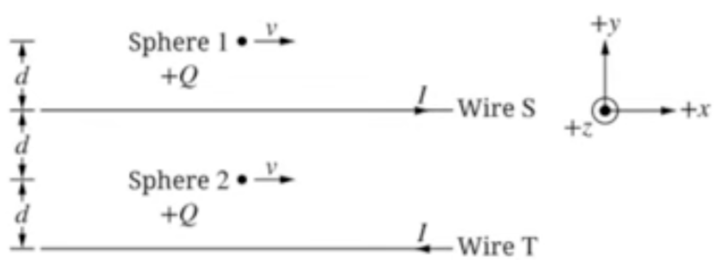
Finally, let us discuss **Ampere's law**. We have that

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{l}} = \mu_0 I_{\text{encl}} \tag{6.26}$$

where I_{encl} is the enclosed charge within the path.

Problem

Long, parallel wires S and T are a distance $2d$ apart. Both wires carry equal currents I, but the currents are in opposite directions. Both wires are parallel to the x-axis. At the instant shown, Sphere 1 is a distance d above wire S. Sphere 2 is a distance d below wire S, and both spheres are moving with speed v in the $+x$-direction. Each sphere has positive charge $+Q$. Gravitational effects are negligible.



1. F_1 is the magnitude of the magnetic force exerted on Sphere 1 due to the currents in wires S and T. F_2 is the magnitude of the magnetic

force exerted on Sphere 2 due to the currents in wires S and T. **Indicate** whether F_2 is greater than, less than, or equal to F_1. **Justify** your answer.

2. **Derive** an expression for the magnitude B_{tot} of the magnetic field at the location of Sphere 2 due to the currents in wires S and T in terms of d, I, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

3. Later, wire T carries current $3I$ in the $+x$-direction. At this instant, Sphere 2 is a distance d below wire S and is moving with speed v in the $+x$-direction. F_{new} is the new magnitude of the magnetic force exerted on Sphere 2 due to the currents in wires S and T. **Indicate** whether F_{new} is greater than, less than, or equal to F_2. Briefly **justify** your answer by referencing your derivation in part 2.

Bonus: Magnetic Fields in Matter

A **magnetic dipole moment** is given by the equation

$$\vec{\mathbf{m}} = I \iint d\vec{\mathbf{A}} = I\vec{\mathbf{A}} \tag{6.27}$$

To find the direction of $\vec{\mathbf{A}}$, which is the vector area of a loop, then curl your fingers in the direction of the current and the thumb will point in the direction of the area vector (once again, use your right hand).

The vector potential of a dipole $\vec{\mathbf{m}}$ is given by (not to be confused with the vector area),

$$\vec{\mathbf{A}}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{\vec{\mathbf{m}} \times \hat{\mathbf{r}}}{r^2} \tag{6.28}$$

When we consider a whole magnetized object, we must construct an integral similar in form to the formula above. Each volume element dV must carry with it a dipole moment $\vec{\mathbf{M}}dV$, so we have that

$$\vec{\mathbf{A}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint \frac{\vec{\mathbf{M}} \times \hat{\mathbf{r}}}{r^2} dV \tag{6.29}$$

Now consider the identity

$$\nabla' \frac{1}{r} = \frac{\hat{\mathbf{r}}}{r^2} \tag{6.30}$$

Now we can say

$$\vec{\mathbf{A}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint \left[\vec{\mathbf{M}} \times \left(\nabla' \frac{1}{r} \right) \right] dV \tag{6.31}$$

If we integrate by parts and use the identity $\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$, we obtain

$$\vec{\mathbf{A}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \left\{ \iiint \frac{1}{r} [\nabla' \times \vec{\mathbf{M}}] dV - \iiint \nabla' \times \left[\frac{\vec{\mathbf{M}}}{r} \right] dV \right\} \tag{6.32}$$

The second term, as it turns out, can be written as a surface integral:

$$\vec{\mathbf{A}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint \frac{1}{r} [\nabla' \times \vec{\mathbf{M}}] dV + \frac{\mu_0}{4\pi} \oiint \frac{1}{r} [\vec{\mathbf{M}} \times d\vec{\mathbf{a}}] \tag{6.33}$$

Yes, the notation is kind of bad here. Just for this equation, the area differential is notated $d\vec{\mathbf{a}}$.

The potential of a volume current (that is, a current in three-dimensional space) is

$$\vec{\mathbf{J}}_b = \nabla \times \vec{\mathbf{M}} \tag{6.34}$$

and the potential of a surface current (two-dimensional space) is

$$\vec{\mathbf{K}}_b = \vec{\mathbf{M}} \times \hat{\mathbf{n}} \tag{6.35}$$

where $\hat{\mathbf{n}}$ is the unit normal vector. Thus, we obtain

$$\vec{\mathbf{A}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint_{\mathcal{V}} \frac{\vec{\mathbf{J}}_b}{r} dV + \frac{\mu_0}{4\pi} \oiint_{\mathcal{S}} \frac{\vec{\mathbf{K}}_b}{r} d\vec{\mathbf{a}} \tag{6.36}$$

What this complicated mathematical expression essentially means is that the vector potential of a magnetized object is equivalent to what would be produced by a volume current defined as above in the material and a surface current defined as above on the boundary. So, instead of integrating dipoles, we look at the bound currents, as explored above, and calculate the field they produce.

Chapter 7

Electromagnetic Induction

If we have a magnet and a coil of wire around the magnet, if the magnet moves relative to the coil, then there is an **induced current**, and the corresponding emf that causes the current is known as the **induced emf**.

Let us observe **Faraday's law**, which states that the induced emf in a closed loop is the negative of the derivative of the magnetic flux with respect to time.

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \tag{7.1}$$

Meanwhile, the total emf in a coil with N turns is

$$\mathcal{E} = -N\frac{d\Phi_B}{dt} \tag{7.2}$$

Lenz's law states that the direction of any magnetic induction effect opposes the cause of the effect. For instance, if the magnetic field is changing in such a way that the vector is increasing, the induced magnetic vector must point in the opposite direction as the original magnetic vector. If the magnetic field is decreasing in magnitude, the induced magnetic vector must point in the same direction as the original.

Moving conductors in a magnetic field can generate what is known as a **motional emf**, such that the magnitude of the emf is

$$\mathcal{E} = vBL \tag{7.3}$$

where v is the conductor speed and L is the conductor length.

Faraday's law can be framed in a different way:

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = -\frac{d\Phi_B}{dt} \tag{7.4}$$

An interesting fact: a change in the current in one circuit can induce a current in a second circuit. Suppose that the magnetic flux through each turn of coil 2, caused by the current i_1 in coil 1, is Φ_{B2}. Furthermore, suppose coil 2 has N_2 turns. The **mutual inductance** of the two coils is

$$M_{21} = \frac{N_2 \Phi_{B2}}{i_1} \tag{7.5}$$

Mutually induced emfs can be found by the following formulas:

$$\mathcal{E}_2 = -M\frac{di_1}{dt}, \quad \mathcal{E}_1 = -M\frac{di_2}{dt} \tag{7.6}$$

Furthermore,

$$M = \frac{N_2 \Phi_{B2}}{i_1} = \frac{N_1 \Phi_{B1}}{i_2} \tag{7.7}$$

The SI unit of mutual inductance is called the **henry**.

A current in a circuit can set up a magnetic field that causes a magnetic flux through the same circuit; this flux changes when the current changes. The induced emf is known as a **self-induced emf**. We define the **self-inductance** of the circuit

$$L = \frac{N\Phi_B}{i} \tag{7.8}$$

and

$$\mathcal{E} = -L\frac{di}{dt} \tag{7.9}$$

Inductance can be thought of as the tendency of an electrical conductor to oppose a change in the electric current flowing through it. A circuit device designed to have a specific inductance is called an **inductor**.

The energy stored in an inductor when the current increases from 0 to I is

$$U = L\int_0^I idi = \frac{1}{2}LI^2 \tag{7.10}$$

The magnetic energy density in a vacuum is

$$u = \frac{B^2}{2\mu_0} \tag{7.11}$$

In a material, it is

$$u = \frac{B^2}{2\mu} \tag{7.12}$$

μ is known as the permeability of the material; μ_0 is sometimes called the vacuum permeability.

An **R-L circuit** includes both a resistor and an inductor. Consider the growth of current with an emf in the series; the formula is

$$i = \frac{\mathcal{E}}{R}(1 - e^{-(R/L)t}) \tag{7.13}$$

and the **time constant** for an R-L circuit is

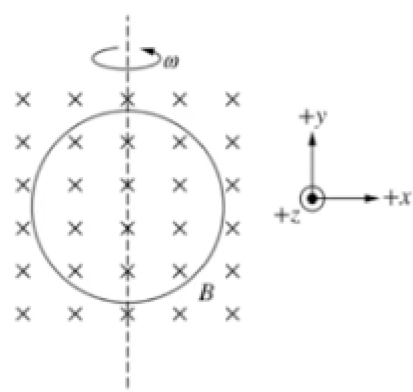
$$\tau = \frac{L}{R} \tag{7.14}$$

When the current has reached the value I_0, and the battery is bypassed, the current begins to decay. The current decays smoothly:

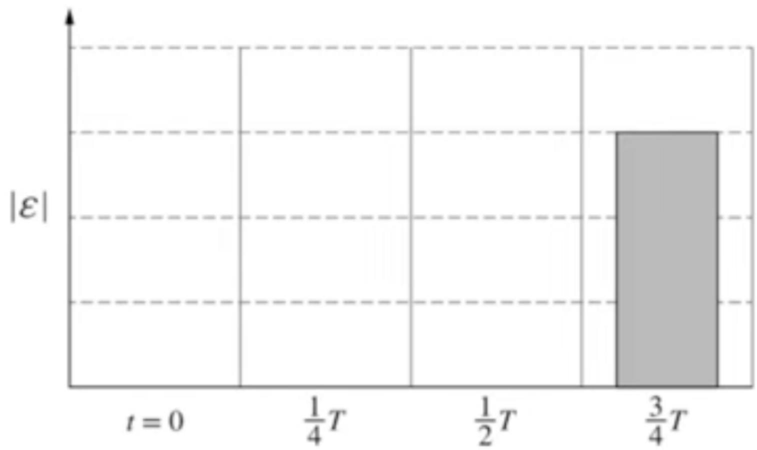
$$i = I_0 e^{-(R/L)t} \tag{7.15}$$

Problem

A rotating, circular, conducting loop of area A and resistance R is in an external uniform magnetic field of magnitude B that is directed in the $-z$-direction. At time $t = 0$, the magnetic field is perpendicular to the plane of the loop. The loop is rotating with constant angular speed ω and period T about the dashed line that is along the diameter of the loop. The value of the magnetic flux through the loop as a function of time t is $\Phi = BA\cos(\omega t)$.



1. The absolute value of the induced emf in the loop is $|\mathcal{E}|$. The partially completed bar chart shows a bar that represents $|\mathcal{E}|$ at $t = \frac{3}{4}T$. **Draw** bars to represent $|\mathcal{E}|$ at times $t = 0$, $\frac{1}{4}T$, and $\frac{1}{2}T$ relative to $|\mathcal{E}|$ shown at $\frac{3}{4}T$. If $|\mathcal{E}| = 0$, **write** a "0" in that column.



2. **Derive** an expression for the maximum induced current in the loop in terms of A, R, B, ω, and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

3. **Sketch** a graph of the instantaneous power P dissipated by the loop as a function of t during the time interval $0 \leq t \leq T$.

4. **Indicate** whether the sketch you drew in part 3 is or is not consistent with the bars that you drew in part 1. Briefly **justify** your answer by referencing the functional dependence between P and $|\mathcal{E}|$.

Bonus: The Continuity Equation

We define the charge in a volume as the volume integral of the charge density:

$$Q(t) = \iiint_{\mathcal{V}} \rho(\mathbf{r}, t) dV \tag{7.16}$$

Now, we know that the current flowing out through the boundary of the volume must be equal to

$$\oint\!\!\!\oint_{\mathcal{S}} \vec{\mathbf{J}} \cdot d\vec{\mathbf{A}} \tag{7.17}$$

so we have that

$$\frac{dQ}{dt} = -\oint\!\!\!\oint_{\mathcal{S}} \vec{\mathbf{J}} \cdot d\vec{\mathbf{A}} \tag{7.18}$$

This gives us, through the divergence theorem,

$$\iiint_{\mathcal{V}} \frac{\partial \rho}{\partial t} dV = -\iiint_{\mathcal{V}} \nabla \cdot \vec{\mathbf{J}} dV \tag{7.19}$$

and thus, we obtain the continuity equation, which is the mathematical formulation of local conservation of charge:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \vec{\mathbf{J}} \tag{7.20}$$

We will not be going over this next theorem, but I do wish to share one particularly important theorem in electrodynamics, which is known as **Poynting's theorem**:

$$\frac{dW}{dt} = -\frac{d}{dt} \iiint_{\mathcal{V}} \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) dV - \frac{1}{\mu_0} \oint\!\!\!\oint_{\mathcal{S}} (\vec{\mathbf{E}} \times \vec{\mathbf{B}}) \cdot d\vec{\mathbf{A}} \tag{7.21}$$

It essentially states that the work done on the charges by the electromagnetic force is equal to the decrease in energy remaining in the fields minus the energy that flowed out through the surface.

Appendix A

Maxwell's Equations

Maxwell's equations in integral form should look familiar. They are:

1. Gauss's law:

$$\oiint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{1}{\epsilon_0} \iiint \rho dV \tag{A.1}$$

2. Gauss's law of magnetism:

$$\oiint \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = 0 \tag{A.2}$$

3. Faraday's law:

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = -\frac{d}{dt} \iint \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} \tag{A.3}$$

4. Ampere's law:

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{l}} = \mu_0 \left(\iint \vec{\mathbf{J}} \cdot d\vec{\mathbf{A}} + \epsilon_0 \frac{d}{dt} \iint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} \right) \tag{A.4}$$

In differential form, they become the following:

$$\nabla \cdot \vec{\mathbf{E}} = \frac{\rho}{\epsilon_0} \tag{A.5}$$

$$\nabla \cdot \vec{\mathbf{B}} = 0 \tag{A.6}$$

$$\nabla \times \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t} \tag{A.7}$$

$$\nabla \times \vec{\mathbf{B}} = \mu_0 \left(\vec{\mathbf{J}} + \epsilon_0 \frac{\partial \vec{\mathbf{E}}}{\partial t} \right) \tag{A.8}$$

Maxwell's Condensed Equations

We begin by writing $\vec{\mathbf{E}}$ not as a vector field, but as a 1-form:

$$E = E_x dx + E_y dy + E_z dz \tag{A.9}$$

and we turn the magnetic field $\vec{\mathbf{B}}$ into a 2-form:

$$B = B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy \tag{A.10}$$

Now, we combine both of these into a 2-form, known as F, in $\mathbb{R}^4$:

$$F = B + E \wedge dt \tag{A.11}$$

where

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix} \tag{A.12}$$

we call this the **electromagnetic field tensor**.

Now, we shall present Maxwell's condensed equations here. The first pair of equations, namely equations A.6 and A.7, becomes simply

$$dF = 0 \tag{A.13}$$

The other pair of equations becomes

$$*d*F = J \tag{A.14}$$

where

$$J = j_1 dx^1 + j_2 dx^2 + j_3 dx^3 - \rho dt \tag{A.15}$$

This formulation is not very useful when it comes to practical calculations; at least, not nearly as useful as the original formulation presented above in the appendix, but this formulation's strengths lie in the field of theoretical physics, where one may employ techniques in differential geometry among other fields of math to study the nature of the universe.